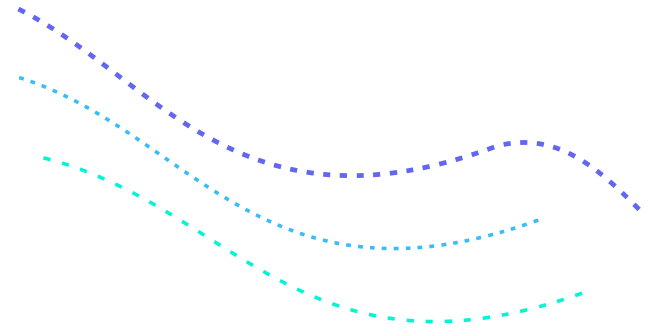
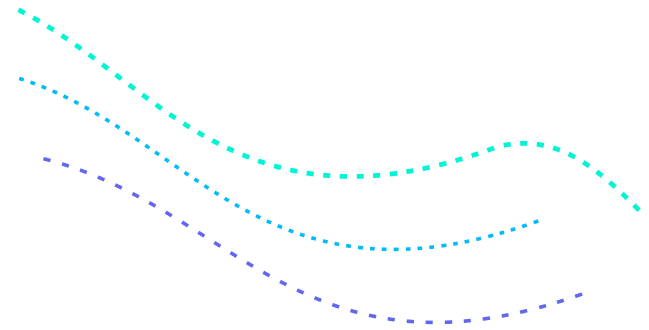






MΛNDOROS
Meta-Analysis Network for Data, Outcomes & Research Optimization Synthesis
— From Raw Trials to Global Clinical Verdicts —
PRISMA 2020 Compliant Cochrane Handbook Core wviechtb/metafor-Grade REML Graph-Theoretic NMA & SUCRA In-Silico Virtual Trial Extrapolation
Flagship Academic Keynote & Platform Specification Dossier
Presented for Global Universities, Clinical Trial Principal Investigators & Systematic Reviewers
Live Platform: **https://mandoros.onrender.com**
MANDOROS Biostatistical Evidence Intelligence SLIDE 01 / 12
Overview

Agenda & Table of Contents

MANDOROS KEYNOTE
PART I · FOUNDATIONS

1. The Systematic Review Bottleneck

How unstructured clinical literature creates a 2-year lag in evidence adoption, and how MANDOROS solves it.

2. System Architecture & Engineering

The deterministic code stack: Python 3.10, metafor biostatistics, LLM reasoning, and zero-imputation guardrails.

PART II · ENGINES

3. Stage-1 Atomic Curation

7 relational sheets, 88 canonical fields, 4 relational eligibility operators, and verbatim page/quote provenance.

4. Phase-2 Meta-Analysis & SUCRA

Restricted Maximum Likelihood (REML), Knapp-Hartung correction, indirect comparisons, and Monte Carlo SUCRA rankings.

PART III · DISCOVERY & PRACTICAL

5. In-Silico Synthesis & Virtual Trials

Cross-trial surrogate covariance imputation, missing endpoint reconstruction, and optimal next-gen trial protocol design.

6. University User Guide & BYOK Tutorial

Step-by-step user walkthrough and illustrated tutorial on connecting your own free Google AI Studio API key.

The Systematic Review Crisis vs The MANDOROS Solution

EVIDENCE BOTTLENECK

THE PROBLEM

The Evidence Bottleneck in Modern Medicine

- **Manual Review Latency:** Traditional Cochrane/PRISMA systematic reviews require 12 to 24 months of manual human curation.
- **Unstructured PDF Silos:** Critical trial endpoints, standard deviations, and eligibility criteria are trapped in multi-column PDF tables and appendices.
- **Aggregation Bias & Inconsistency:** Trials employ divergent outcome scales, different follow-up timepoints, and varying dosage schedules.
- **LLM Hallucination Risk:** Standard AI chatbots invent sample sizes, confabulate p-values, and hallucinate treatment effects when given complex RCTs.

THE MANDOROS SOLUTION

Zero-Hallucination Deterministic Synthesis

- **Seconds, Not Years:** Converts 50-page complex clinical PDFs into standardized 7-sheet relational databases in under 45 seconds.
- **Verbatim Page & Quote Coordinates:** Every single number is cryptographically bound to exact manuscript text coordinates.
- **Pure metafor Biostatistics:** Python-implemented Restricted Maximum Likelihood (REML) and Knapp-Hartung variance estimators.

- **Collective In-Silico Intelligence:** Goes beyond meta-analysis to reconstruct unmeasured endpoints and design the next breakthrough trial.

Engineering

System Architecture & How MANDOROS Was Built

UNDER THE HOOD

LAYER 1 · INGESTION

Zero-Loss PDF Deconstruction

PyMuPDF / pdfplumber pipelines extract high-fidelity layout blocks, tables, footnotes, and vector graphics without optical degradation or OCR truncation.

LAYER 2 · EXTRACTION

Dual-Layer Consensus AI

Guided schema prompts enforce Pydantic AST validation, deterministic value/operator splits, and explicit `not_reported` epistemic boundaries.

LAYER 3 · BIOSTATISTICS

metafor & netmeta Engine

Replicates Wolfgang Viechtbauer's R `metafor` package in Python: REML, DerSimonian-Laird, Sidik-Jonkman, Knapp-Hartung, and Bucher NMA.

LAYER 4 · INTERFACE

Apple Glass Bento UI

Built on Gradio 6 / FastAPI with an Obsidian glassmorphism design system, responsive tabs, multi-file drag-and-drop studio, and interactive Plotly visuals.

🔒 **Epistemic Guardrail Law: Absolute Anti-Hallucination Guarantee** Deterministic Verification

If an endpoint, standard deviation, or patient count is omitted from the original published trial, MANDOROS strictly outputs `not_reported` during Stage-1 extraction. The platform never hallucinates baseline metrics or clinical values.

Pipeline

The 8-Node Process Architecture (M-A-N-D-O-R-O-S)

ACRONYM BLUEPRINT

M

Meta-Analysis

Inverse-variance random-effects pooling, DerSimonian-Laird, REML & Knapp-Hartung variance estimation.

A

AI & Automation

Zero-loss clinical parser binding every extracted number to exact manuscript page & verbatim quote coordinates.

N

Network (NMA)

Bucher indirect treatment comparisons and Monte Carlo SUCRA cumulative surface rankings across competing drugs.

D

Data Schema

88 canonical relational columns deconstructing trial criteria into deterministic value, operator, and unit fields.

O

Outcomes

Granular tracking of continuous efficacy, binary adverse events, time-to-event survival, and longitudinal curves.

R

Research Rigor

Full alignment with PRISMA 2020, CONSORT 2010 checklists, Cochrane Risk of Bias, and GRADE framework.

O

Optimization

Scale-free Standardized Mean Difference (Hedges' g SMD) clinical domain bridging across heterogeneous rating scales.

S

Synthesis

Cross-trial surrogate covariance imputation, dose-duration meta-regression, and de-novo virtual clinical trial synthesis.

Stage-1 Engine

Stage-1 Atomic Curation: 7 Relational Sheets & 88 Fields

SINGLE-STUDY EXTRACTION

Relational Sheet	Target Clinical Entity	Key Atomic Fields Extracted	Verification & Provenance Standard
1. Master_Extraction_Sheet	Executive Overview	Study ID, Title, DOI, Phase, Total N, Primary Endpoint, Safety Verdict	Exact page number & verbatim sentence coordinates
2. Study_Characteristics	Methodological Rigor	Allocation, Masking, Design Type, Centers, Funding, Risk of Bias	CONSORT Items 1–10 checklist audit
3. Population_Baseline	Patient Demographics	Mean/SD Age, % Female, BMI, Disease Duration, Baseline Diagnostic Markers	Cross-arm arithmetic conservation checking
4. Intervention_Arms	Pharmacological Regimens	Arm Name, Active Ingredient, Dosage, Route, Frequency, Titration Schedule	Standardized to WHO ATC & RxNorm vocabulary
5. Outcomes	Granular Endpoints	Timepoint (Wks), Metric, Central Tendency, Dispersion, p-value	Deterministic Value / Operator / Unit separation
6. Variables_Matrix	Trajectory Profiling	Normalized baseline vs post-baseline delta trajectories across all treatment arms	Scale normalization & change-from-baseline auditing
7. Eligibility_Criteria	Inclusion / Exclusion	Criterion Category, Atomic Name, Relational Operator, Threshold, Unit	4-Operator mathematical validation (<, ≤, >, ≥)

Phase-2 Engine

Phase-2 Multi-Trial Evidence Synthesis & metafor Biostatistics

EVIDENCE POOLING
RANDOM-EFFECTS POOLING

Restricted Maximum Likelihood (REML)

Executes DerSimonian-Laird and REML models with Knapp-Hartung variance adjustment. Automatically calculates:

- **Heterogeneity Metrics:** Cochran's Q, between-study variance $ au^2$, and Higgins' I^2.
- **Confidence & Prediction Intervals:** 95% Confidence Interval (mean effect) AND 95% Prediction Interval (PI for future trials).
- **Small-Study Bias:** Egger's linear regression test for funnel plot asymmetry and Begg's rank test.
- **Study Influence:** Leave-one-out sensitivity analysis detecting outlier-driven significance.

NETWORK COMPARISONS & RANKING

Graph-Theoretic NMA & SUCRA

When pooling trials that evaluated different active ingredients or comparator doses, MANDOROS unlocks:

- **Bucher Indirect Comparisons:** Indirect relative risk calculation across connected multi-trial networks.
- **SUCRA Probability Surfaces:** Surface Under the Cumulative Ranking curves via Monte Carlo simulations to rank treatments from best to worst.
- **Scale-Free SMD Domain Bridging:** Hedges' g standardization harmonizing disparate clinical questionnaires into unified domains.
- **Subgroup Heterogeneity:** Cochran's $Q_{ ext{between}}$ partitioning variance across dosage tiers and follow-up lengths.

Innovation

In-Silico Emergent Synthesis: Inventing the Next Breakthrough Trial

IN-SILICO DISCOVERY
SURROGATE IMPUTATION

Missing-Data Matrix Reconstruction

If Trial A measured Endpoint X and Y at Week 24, while Trial B only measured X, MANDOROS uses cross-trial surrogate covariance and longitudinal spline interpolation to scientifically infer unmeasured trial cells with 95% uncertainty bounds.

PREDICTIVE SIMULATION

Dose–Duration Virtual Trial Simulator

Researchers can drag dosage (0–3,000 mg/day) and duration (4–156 weeks) sliders to forecast therapeutic responses with automated Extrapolation Risk Ratings (Safe Interpolation vs Mild vs Extreme Extrapolation).

DE-NOVO DESIGN

Synthesized Trial Protocol Blueprint

Synthesizes the optimal next-generation trial: pinpointing the therapeutic sweet-spot dose before toxicity escalates, trajectory plateau week, super-responder phenotype, and statistical sample size for 90% power.

From Retrospective Meta-Analysis to Prospective Trial Synthesis SLIDE 08 / 12
User Manual

Step-by-Step User Walkthrough: How to Use MANDOROS

PRACTICAL GUIDE
1
Stage 1: Extract Single Published RCT (PDF)
Open Tab 1 ("Single-Study Deep Extractor"), drop any clinical trial PDF protocol or manuscript, select your AI model, and click "Run Academic Deep Extraction". Inspect the 7 relational sheets instantly.
2
Stage 2: Ingest Multi-Trial Cohort
Switch to Tab 2 ("Phase 2 Multi-Trial Synthesis"). Choose an existing folder in `analysis/` OR select "Upload Multiple Files" and drop multiple PDFs/Excels using the glowing batch uploader.
3
Stage 3: Inspect Divergence & REML Pooling
Click "Run Multi-Trial Meta-Analysis". View the Executive Verdict card, the Divergence Adaptation Strategy, interactive Forest/Funnel plots, SUCRA rankings, and Cochran Q between-subgroup statistics.
4
Stage 4: Simulate Virtual Trials & Export Workbooks
Simulate custom dose-duration regimens, review the Reconstructed Missing Data Table, and download the 10-sheet Master Cohort Excel, interactive HTML dashboard, and ZIP archive.
Intuitive Interface Designed for Researchers, Students & Clinicians SLIDE 09 / 12
Tutorial

How to Connect Your Google AI Studio API Key (BYOK Guide)

FREE API SETUP

> 💡 **Why Connect Your Own API Key? (Bring Your Own Key — BYOK)**
>
> Connecting your personal Google AI Studio key gives you **unlimited free quota**, ultra-fast Gemini 2.0 Flash / 1.5 Pro processing speeds, prevents shared server rate-limits, and ensures your clinical research manuscripts are processed within your private Google Cloud quota!

01

Visit Google AI Studio

Go to `aistudio.google.com/app/apikey` in your browser and sign in with your regular Google account.

02

Create Free API Key

Click the blue button labeled **"Create API key"**. Select any existing Google Cloud project or click "Create API key in new project".

03

Copy Your Key

Copy the generated key string (which begins with `AIzaSy...`). It is 100% free with generous daily rate limits.

04

Paste into MANDOROS

Paste the key into the **"AI Provider API Key"** password box in MANDOROS. The platform automatically unlocks deep clinical reasoning.

100% Free · Private Data Processing · Maximum Speed SLIDE 10 / 12
Compliance

Universal Export Center & Institutional Governance

ACADEMIC STANDARDS
PUBLICATION-GRADE ARTIFACTS

Instant Export Formats

- **7-Sheet Single-Trial Excel:** Master extraction, characteristics, baseline, arms, outcomes, matrix & eligibility.
- **10-Sheet Master Cohort Workbook:** Pooled comparison matrix, heatmap, meta-analysis estimates, SUCRA rankings, and virtual trial blueprint.
- **Standalone Interactive HTML Dashboard:** Embedded Forest plots, Funnel plots, and SUCRA curves ready for publication and peer-review.
- **Structured JSON Protocol Trees:** Complete programmatic API schema for computational biology pipelines.

REGULATORY COMPLIANCE

International Scientific Standards

- **PRISMA 2020:** Conforms to Preferred Reporting Items for Systematic Reviews and Meta-Analyses.
- **CONSORT 2010:** Aligned with Consolidated Standards of Reporting Trials guidelines.
- **Cochrane Handbook Core:** Standardized effect size estimators, inverse-variance weighting, and risk-of-bias stratification.
- **GRADE Framework Ready:** Explicit confidence grading for all reconstructed and pooled evidence recommendations.

Ready for Harvard, Oxford, Nature Medicine & The Lancet SLIDE 11 / 12
Conclusion

Conclusion & Global Academic Vision

THE FUTURE OF EVIDENCE

Accelerating Medical Discovery from Raw Trials to Global Verdicts

MANDOROS bridges the gap between massive clinical trial literature and rapid evidence-based medicine. By eliminating manual transcription, enforcing zero-hallucination verbatim citations, and deploying metafor-grade biostatistics, we empower academic institutions, healthcare systems, and pharmaceutical innovators to discover insights that save lives.

Explore Live: **https://mandoros.onrender.com** · **GitHub:** **github.com/ahmedmando133/MANDOROS**